

Detecting departure and destination aerodromes from ADS-B

Version 6 — the consolidated study, and what to run in production

EUROCONTROL Performance Review Unit

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Summary

OPDI derives a flight’s departure and destination aerodrome from ADS-B state vectors alone. This page consolidates six versions of that work into one account: what the three candidate methods are, how each behaves across its whole parameter space, which settings are best for departures and for arrivals, what the ideas that did *not* work were, and what the pipeline should ship.

The short version is in three parts.

The two roles want different algorithms. Departures are best served by `endpoint` — take the aerodrome nearest the first fix, but only commit when the fix is close and low, and abstain otherwise. Arrivals are best served by `trend`, the algorithm already in production. That split holds at every exchange rate tested.

Both roles improve. The recommended configuration — `endpoint` for departures, exact-distance `trend` for arrivals — scores **133,807** against production’s **123,659** on a three-day sample of 95,116 flights: a gain of **+10,148**, every figure from a real pipeline run.

Arrivals improve too, but only after a defect in the pipeline was fixed — and that is the result worth carrying away. Sweeping `trend` said its never-tuned flight-level gate should move from FL40 to FL60, worth over a thousand flights. Run through the pipeline, the same change *lost* ground.

The cause was not the tuning. The sweep ranked candidate aerodromes by exact distance; the pipeline kept only the candidates at the minimum **H3 ring count** — an integer stepping about 5.2 km at resolution 7 — and measured only among those. An aerodrome a kilometre further out was discarded before its distance was ever computed. Raising the flight-level gate admits samples from higher up, where more aerodromes fall in the same ring, so the coarser rule had more work to do exactly where it was least able to do it.

The sweep was not over-estimating. It was modelling a better algorithm than production implemented. With ranking on exact distance, every tuned arrival value gains ground, and the pipeline’s own optimum turns out to be the one the sweep had been pointing at all along.

Reading this page

Every number below is an inline expression over a CSV in `data/`, and every one of those CSVs is produced by a job in `benchmarks/regenerate_v6.py` — which this page runs before drawing anything. Nothing is typed into the prose, and nothing is read that the checked-out code did not produce. `?@sec-provenance` lists the command and commit behind each figure. Where this page disagrees with versions 1–5, it says so and gives the reason; those versions stay published unchanged.

1 How the metrics work

Five quantities are reported for every setting, for each role separately. Four of them are unavoidable and the fifth is a judgement, so it is worth being precise about all five before any of them are used to argue for a parameter.

Write N for the number of ground-truth flights in the sample, a for the number the method *answers* (names an aerodrome, or explicitly says out-of-area), c for the number it answers **correctly**, and $w = a - c$ for the number it answers **wrongly**.

Coverage is a/N : the share of flights the method is willing to commit on. A method that abstains everywhere has coverage zero and cannot be wrong.

Accuracy is c/a : of the flights it committed on, the share it got right. Accuracy says nothing about how often the method declined to answer, so a method can buy accuracy simply by abstaining more.

Correct (c) and **wrong** (w) are the raw counts. They matter because coverage and accuracy generally move in opposite directions — every extra answer a method gives is one it was previously unsure about, so raising coverage tends to lower accuracy — and neither ratio says whether a particular exchange was worth making. The counts do.

Score is $c - kw$. It exists to make the exchange rate explicit: a change to a parameter is worth taking if and only if it wins more than k correct answers for each wrong one it introduces.

The value of k is where judgement enters. This study uses $k = 2$ by default, on the following argument. A **silence** — an abstention — corrupts one movement count, and it is *visible*: it arrives downstream as an explicit null that any consumer can filter on. A **wrong aerodrome** corrupts *two* movement counts, the one it invents and the one it omits, and it is *invisible*: nothing downstream can distinguish it from a correct answer. Two errors for one, plus undetectability, is the case for $k = 2$.

That is an argument, not a measurement, and rankings do move with k . So every headline choice on this page is also scored at $k = 1$ and $k = 3$, and where the ranking changes, the page says so.

1.1 Out-of-area, and why the headline numbers are decomposed

Roughly one flight in six in the sample genuinely begins or ends outside the observed area. Such a flight has no nameable aerodrome, and the correct answer is the literal label **OOA**. This creates a way to score well for a bad reason: a method that labels liberally out-of-area collects those flights for free.

So each side is also reported three further ways — out-of-area recall and precision, which show whether the free wins are being taken honestly, and in-area coverage and accuracy, which restrict to flights whose truth is a real aerodrome and are the numbers to read when comparing naming rules.

Table 1: The reported quantities.

Quantity	Definition	Reads as
Coverage	a/N	willingness to commit
Accuracy	c/a	reliability when committed
Correct	c	flights named right
Wrong	$w = a - c$	flights named wrong
Score	$c - kw$	net value at exchange rate k
OOA recall	OOA hits / OOA in truth	does it recognise what it cannot name?
OOA precision	OOA hits / OOA predicted	is it over-using the OOA label?
In-area coverage	answered / truth is a real aerodrome	commitment with the free wins removed
In-area accuracy	correct / answered, truth a real aerodrome	reliability with the free wins removed

2 The sample

The 2025 period is 5-7 June 2025, three consecutive days, giving **95,116 ground-truth flights** after alignment with the Network Manager reference.

Ground truth is the EUROCONTROL Network Manager flight table, joined to ADS-B tracks on aircraft address, callsign and date. The join key is **AIRCRAFT_ADDRESS**, which is the **icao24** of the ADS-B message.

3 The three methods

3.1 trend — what production runs today

For every state vector below a flight-level cap and within a radius of an aerodrome, **trend** asks whether smoothed barometric altitude is rising or falling. Each vector casts a vote. If climbing votes exceed descending votes by more than a margin, that aerodrome is the departure; if the reverse, the destination. The nearest surviving candidate wins.

Two of its four constants had never been swept before this study: the flight-level cap, fixed at FL40, and the detection radius, fixed at 30 NM.

3.2 nearest — the *traffic* library’s rule

The reference implementation in the `traffic` library takes the aerodrome nearest the first and last point of the trajectory, unconditionally. It never abstains, so its coverage is total and its accuracy is whatever the geometry gives. It is included as the external baseline.

3.3 endpoint — nearest, with abstention

`endpoint` is `nearest` plus two gates: the endpoint must lie within a radius of the aerodrome, and either on the ground or below a height above the field. Failing both, the method abstains rather than guessing. A scheduled-service penalty breaks ties toward aerodromes with commercial traffic.

4 Trend: the full parameter sweep

4.1 The untuned baseline

- **ADEP** at the production setting (FL40, margin 4, 30 NM, no penalty): coverage 68.35%, accuracy 98.73%, 64,188 correct, 826 wrong, score 62,536.
- **ADES** at the production setting (FL40, margin 4, 30 NM, no penalty): coverage 67.38%, accuracy 97.52%, 62,500 correct, 1,589 wrong, score 59,322.

4.2 The best setting found

- **ADEP**: FL120, margin 0, 20 NM, penalty 10 NM — coverage 77.43%, accuracy 98.58%, 72,607 correct, 1,045 wrong, score 70,517 (+7,981 against the untuned baseline).
- **ADES**: FL60, margin 2, 20 NM, penalty 10 NM — coverage 69.50%, accuracy 97.44%, 64,416 correct, 1,689 wrong, score 61,038 (+1,716 against the untuned baseline).

4.3 The full trend grid

`trend` has three geometric parameters, so the sweep is a cube and cannot be printed as one table. It is sliced instead: for each role, score by flight-level cap against radius at that role’s best vote margin, then the margin’s own profile, then the penalty. Every cell of the cube is in `data/trend_sweep_2025.csv`.

ADEP — score at $k = 2$, by flight-level cap (rows) and radius in NM (columns), at the winning vote margin of 0

FL cap	5	10	15	20	30	40	60	80
20	56,065	55,966	55,909	55,855	55,725	55,663	55,403	54,601
30	62,657	62,897	62,821	62,746	62,588	62,520	62,198	61,456
40	64,619	65,913	65,830	65,746	65,543	65,451	65,108	64,229
60	65,139	67,927	67,959	67,872	67,621	67,498	67,123	66,283
80	64,942	68,768	69,395	69,347	69,082	68,910	68,487	67,628
100	64,650	68,797	70,080	70,246	69,966	69,723	69,234	68,350
120	64,189	68,383	70,037	70,517	70,339	70,066	69,543	68,634
150	62,892	67,206	68,925	69,464	69,586	69,326	68,727	67,826

FL cap	5	10	15	20	30	40	60	80
200	59,862	63,905	65,286	65,617	65,739	65,530	64,855	63,936

ADES — score at $k = 2$, by flight-level cap (rows) and radius in NM (columns), at the winning vote margin of 2

FL cap	5	10	15	20	30	40	60	80
20	58,185	58,844	58,612	58,471	58,316	58,213	58,064	57,996
30	58,522	60,169	60,076	59,944	59,781	59,666	59,532	59,448
40	58,435	60,290	60,675	60,720	60,573	60,450	60,306	60,209
60	57,654	59,020	59,740	61,038	61,023	60,867	60,652	60,519
80	56,703	58,043	58,717	60,259	60,476	60,292	60,027	59,838
100	54,684	55,990	56,625	58,100	58,595	58,482	58,176	57,916
120	52,961	54,175	54,842	56,035	56,532	56,423	56,086	55,794
150	51,410	52,237	52,637	53,639	53,887	53,788	53,428	53,112
200	48,982	49,205	49,106	49,642	49,346	49,066	48,714	48,383

The vote margin, at each role’s winning cap and radius:

Role	Margin	Coverage	Accuracy	Correct	Wrong	Score
ADEP	0	77.43%	98.58%	72,607	1,045	70,517
ADEP	2	77.18%	98.64%	72,409	997	70,415
ADEP	4	76.98%	98.68%	72,251	968	70,315
ADEP	8	76.32%	98.72%	71,665	930	69,805
ADEP	16	74.86%	98.78%	70,333	868	68,597
ADES	0	69.65%	97.36%	64,500	1,752	60,996
ADES	2	69.50%	97.44%	64,416	1,689	61,038
ADES	4	69.11%	97.52%	64,105	1,631	60,843
ADES	8	68.60%	97.59%	63,676	1,574	60,528
ADES	16	67.36%	97.69%	62,591	1,477	59,637

The scheduled-service penalty applied to trend, which production does not do at all, swept at each role’s winning geometry:

Role	Penalty (NM)	Coverage	Accuracy	Correct	Wrong	Score
ADEP	0	77.43%	96.37%	70,977	2,675	65,627
ADEP	5	77.43%	98.39%	72,463	1,189	70,085
ADEP	10	77.43%	98.58%	72,607	1,045	70,517
ADEP	15	77.43%	98.44%	72,504	1,148	70,208
ADEP	20	77.43%	98.29%	72,391	1,261	69,869
ADEP	30	77.43%	98.29%	72,391	1,261	69,869
ADES	0	69.50%	95.34%	63,025	3,080	56,865
ADES	5	69.50%	97.14%	64,216	1,889	60,438
ADES	10	69.50%	97.44%	64,416	1,689	61,038
ADES	15	69.50%	97.37%	64,369	1,736	60,897
ADES	20	69.50%	97.28%	64,306	1,799	60,708
ADES	30	69.50%	97.28%	64,306	1,799	60,708

Coverage is constant down each block and only accuracy moves, which is the signature of a pure re-ranking: the penalty changes which aerodrome is named, never whether one is.

5 Endpoint: the full parameter sweep

endpoint has two geometric parameters. The **radius** is how close to an aerodrome the trajectory's first or last point must be; the **height** is how far above field elevation it may be. Failing either, the method abstains.

The grid below is the whole sweep, scored for each role. The height row labelled *unrestricted* removes the height gate entirely, which collapses **endpoint** into **nearest** within the radius — so the last row of each table is the cost of having no abstention at all.

ADEP — score at $k = 2$, by radius (NM, columns) and height (rows)

height (ft)	5	10	20	30	40	60	80	100	110
0	28,778	28,709	28,684	28,660	28,655	28,643	28,643	28,643	28,643
500	39,506	39,417	39,358	39,324	39,319	39,307	39,307	39,307	39,307
1,000	52,441	52,334	52,249	52,203	52,199	52,177	52,177	52,177	52,177
2,000	62,124	61,996	61,893	61,828	61,818	61,770	61,766	61,766	61,766
5,000	66,124	68,515	68,366	68,259	68,184	68,070	68,043	68,043	68,043
8,000	66,271	70,079	70,668	70,477	70,368	70,250	70,215	70,213	70,213
10,000	66,246	70,288	71,775	71,591	71,437	71,289	71,242	71,240	71,238
15,000	66,220	70,295	72,591	72,755	72,541	72,261	72,202	72,196	72,194
20,000	66,187	70,158	72,258	72,370	72,144	71,758	71,666	71,648	71,646
25,000	66,142	69,990	71,782	71,470	70,623	69,819	69,656	69,629	69,623
30,000	66,111	69,867	71,159	70,179	68,767	67,474	67,261	67,156	67,149
35,000	66,025	69,221	69,355	66,954	64,171	61,585	61,218	60,979	60,936
40,000	65,990	68,914	68,206	65,014	61,108	57,105	56,349	55,817	55,598
unrestricted	65,993	68,905	68,171	64,951	61,010	56,972	56,200	55,648	55,419

ADES — score at $k = 2$, by radius (NM, columns) and height (rows)

height (ft)	5	10	20	30	40	60	80	100	110
0	32,816	32,719	32,597	32,449	32,372	32,282	32,266	32,262	32,262
500	48,233	48,127	47,950	47,785	47,706	47,611	47,595	47,587	47,587
1,000	52,528	52,397	52,172	51,975	51,881	51,770	51,746	51,738	51,738
2,000	53,566	54,560	54,321	54,088	53,959	53,833	53,791	53,781	53,781
5,000	53,524	55,632	57,190	56,955	56,735	56,523	56,420	56,410	56,410
8,000	53,607	55,658	57,703	57,797	57,493	57,250	57,119	57,105	57,105
10,000	53,616	55,606	57,663	57,969	57,684	57,414	57,275	57,261	57,261
15,000	53,640	55,513	57,133	57,116	56,759	56,420	56,267	56,251	56,249
20,000	53,697	55,406	56,381	55,330	54,666	54,218	54,039	53,999	53,997
25,000	53,818	55,447	55,946	54,362	53,406	52,565	52,286	52,228	52,222
30,000	53,900	55,400	55,457	53,361	51,914	50,641	50,103	50,010	49,995
35,000	54,022	54,821	53,200	49,348	46,467	43,182	41,661	41,019	40,805
40,000	54,808	55,179	52,368	47,092	42,991	37,190	34,908	33,249	32,673
unrestricted	54,961	55,315	52,464	47,110	42,854	36,840	34,520	32,809	32,217

5.1 Where each role's optimum sits

- **ADEP**: 30 NM, 15,000 ft — coverage 80.96%, accuracy 98.16%, 75,587 correct, 1,416 wrong, score 72,755.
- **ADES**: 30 NM, 10,000 ft — coverage 67.23%, accuracy 96.89%, 61,951 correct, 1,991 wrong, score 57,969.

Both optima are interior to the grid in both parameters, so neither is an artefact of where the sweep stopped.

6 The scheduled-service penalty

Both methods choose among surviving candidates by distance. Near a major aerodrome that often means choosing between the airport itself and a small airfield a few miles away that almost never sees commercial traffic. The penalty adds a fixed distance to every candidate without scheduled service, biasing the tie-break toward the aerodrome more likely to be the real one.

It is a pure re-ranking: it changes *which* aerodrome is named, never *whether* one is named. So coverage is flat across the whole sweep and only accuracy moves — which is exactly what makes it cheap.

Endpoint, at 40 NM and 15,000 ft:

Role	Penalty (NM)	Coverage	Accuracy	Correct	Wrong	Score
ADEP	0	81.20%	97.31%	75,158	2,077	71,004
ADEP	5	81.20%	97.72%	75,470	1,761	71,948
ADEP	10	81.20%	97.98%	75,667	1,563	72,541
ADEP	15	81.19%	97.90%	75,610	1,619	72,372
ADEP	20	81.20%	97.68%	75,444	1,793	71,858
ADEP	30	81.19%	97.59%	75,365	1,864	71,637
ADES	0	70.00%	93.85%	62,487	4,092	54,303
ADES	5	69.99%	94.31%	62,786	3,787	55,212
ADES	10	69.97%	95.10%	63,287	3,264	56,759
ADES	15	69.92%	94.88%	63,105	3,404	56,297
ADES	20	69.91%	94.80%	63,037	3,456	56,125
ADES	30	69.82%	94.84%	62,985	3,425	56,135

7 How much the exchange rate matters

k is a judgement, so the honest thing is to show what it decides. The table below re-runs the argmax at $k = 1, 2$ and 3 for both methods and both roles.

k	Role	Best trend	trend score	Best endpoint	endpoint score	Winner
1	ADEP	FL120, margin 0, 20 NM, pen 10	71,562	40 NM, 20,000 ft, pen 10	74,233	endpoint
1	ADES	FL80, margin 2, 30 NM, pen 10	62,826	40 NM, 15,000 ft, pen 10	60,023	trend

k	Role	Best trend	trend score	Best endpoint	endpoint score	Winner
2	ADEP	FL120, margin 0, 20 NM, pen 10	70,517	30 NM, 15,000 ft, pen 10	72,755	endpoint
2	ADES	FL60, margin 2, 20 NM, pen 10	61,038	30 NM, 10,000 ft, pen 10	57,969	trend
3	ADEP	FL120, margin 0, 20 NM, pen 10	69,472	20 NM, 15,000 ft, pen 10	71,400	endpoint
3	ADES	FL40, margin 2, 15 NM, pen 10	59,482	20 NM, 8,000 ft, pen 10	56,368	trend

The **choice of method is the same at every k tried**: departures want **endpoint** and arrivals want **trend** whether a wrong answer is priced at one silence, two or three. That is the most robust finding on this page, and it does not depend on the judgement that fixes k .

The parameter *values* are less stable than the method choice, and in a direction that makes sense: a higher k prices wrong answers more dearly, so the argmax moves toward tighter gates that answer less often and are right more often when they do. Where a headline value moves with k , the justification table records it.

8 Negative results, re-measured

Ideas that were tried and did not work. All are re-run here against current code rather than quoted from the version that first reported them.

8.1 The approach cone

The box rule treats radius and height as independent, and they are not: an aircraft descending toward an aerodrome trades height for distance at roughly a constant rate. A 3° glideslope is 318 ft/NM and the “three times the flight level” rule of thumb about 333 ft/NM, so the endpoints that genuinely belong to an aerodrome should lie in a *cone*, not a box.

The two shapes disagree in both corners. Close and high is inside the box and outside the cone — an aircraft at 15,000 ft overhead is overflying, not arriving. Far and low is outside the box and inside the cone — an arrival whose reception dies at 8,000 ft and 30 NM is on profile.

Replacing the box with a cone is worse for both roles, at every slope swept.

- **ADEP**: best cone (1000 ft/NM, 40 NM) scores 65,735 against the box’s 72,755, $-7,020$. It is worse on both counts — fewer correct *and* more wrong, so no exchange rate rescues it.
- **ADES**: best cone (500 ft/NM, 40 NM) scores 57,614 against the box’s 57,969, -355 . It is more restrictive, giving up 443 correct to avoid 44 wrong — a trade worth taking only if a wrong answer cost more than 10 silences.

The cone loses for both roles at every slope swept, and the reason is the one that sank every other rate-based idea in this study: the OpenSky archive carries positions forward between updates, so a *position* is trustworthy whenever it appears, while anything derived by dividing by elapsed time is not. A glideslope is a rate in disguise.

8.2 The bearing test

Version 5 proposed asking whether the trajectory *points at* the aerodrome: take the track’s course near its endpoint, take the bearing from that endpoint to the candidate, and require the two to agree within some angle. It was the first signal found that could add answers the abstention refused, and it was proposed in four forms — **rescue** (answer anyway when aligned), **veto** (require alignment even when the gate accepts), **replace** (alignment instead of the gate) and **rerank** (choose *which* aerodrome by alignment).

All four are re-measured here against the current **endpoint** baseline.

Variant	ADEP cover- age	ADEP accu- racy	ADEP score	Δ	ADES cover- age	ADES accu- racy	ADES score	Δ
base	80.96%	98.16%	72,755	—	69.10%	95.63%	57,116	—
rescue: gate OR align $\leq$ 0.1	81.06%	98.05%	72,585	−170	69.66%	95.52%	57,358	+242
rescue: gate OR align $\leq$ 0.25	81.13%	98.00%	72,528	−227	69.85%	95.44%	57,346	+230
rescue: gate OR align $\leq$ 1.0	81.49%	97.73%	72,224	−531	70.17%	95.23%	57,203	+87
rescue: gate OR align $\leq$ 3.0	82.48%	96.91%	71,180	−1,575	70.75%	94.93%	57,048	−68
veto: gate AND align $\leq$ 0.1	2.66%	93.40%	2,029	−70,726	11.67%	97.30%	10,204	−46,912
veto: gate AND align $\leq$ 0.25	5.03%	96.11%	4,222	−68,533	13.58%	97.57%	11,974	−45,142
veto: gate AND align $\leq$ 1.0	17.15%	98.46%	15,555	−57,200	16.62%	97.86%	14,793	−42,323
veto: gate AND align $\leq$ 3.0	39.83%	99.04%	36,789	−35,966	22.78%	98.17%	20,477	−36,639
replace: align $\leq$ 0.1 only	2.76%	90.26%	1,859	−70,896	12.23%	96.59%	10,446	−46,670
replace: align $\leq$ 0.25 only	5.20%	93.59%	3,995	−68,760	14.33%	96.51%	12,204	−44,912
replace: align $\leq$ 1.0 only	17.68%	96.45%	15,024	−57,731	17.69%	96.14%	14,880	−42,236
replace: align $\leq$ 3.0 only	41.36%	96.51%	35,214	−37,541	24.42%	95.95%	20,409	−36,707
rerank by alignment, then gate	51.52%	86.61%	29,317	−43,438	35.81%	86.50%	20,262	−36,854
rerank, gate OR align $\leq$ 0.1	52.72%	84.74%	27,189	−45,566	37.42%	84.44%	18,982	−38,134
rerank, gate OR align $\leq$ 0.25	54.32%	82.45%	24,461	−48,294	38.73%	82.22%	17,188	−39,928

The best variant of any kind is **rescue: gate OR align $\leq$ 0.1**, worth +242 on arrivals — and −170 on departures, which is the role **endpoint** is actually used for. Nothing here is worth adopting.

The three variants that let bearing *remove* or *choose* answers are not close: the best of them gives up $-35,966$ on departures. Alignment is far too blunt to name an aerodrome — at 40 NM a runway subtends a couple of degrees, but so does every other aerodrome behind it on the same radial, and a track on final approach points at all of them equally.

This supersedes version 5's recommendation. That version scored the bearing rescue against **endpoint** alone and never against **trend**, which was already better than both — the comparison that mattered was the one not made.

8.3 Bearing applied to trend

The bearing variants above all modify **endpoint**. **trend** is the algorithm that ships for arrivals, and it fails differently — it has no gate, so it almost always answers, and its errors are misnamings rather than silences. The variant that suits it is therefore a *rerank* or a *tie-break*, not a rescue, and its candidate set is already narrowed by the vote rule, so alignment is asked a much easier question than on the endpoint side.

Not yet regenerated.

8.4 The vertical measures

Whether a track is climbing or descending can be read from the broadcast vertical rate, from a two-point altitude slope, or by least squares over the whole window. The three disagree, and version 5 could not say which was wrong.

Not yet regenerated.

8.5 State-vector decimation

OPDI thins the 1 Hz feed to roughly 5 s. The production rule is a fixed-phase **event_time % 5** filter; the alternative keeps the newest sample in each 5 s bucket, which cannot drop a sample where the modulo rule would.

Not yet regenerated.

9 What the pipeline actually produces

Every number so far comes from a sweep harness, not from **src/opdi**. The trend sweep re-applies the vote rule to a cached vote table and the endpoint sweeps filter and re-rank a cached candidate table — which is what makes hundreds of parameter cells affordable, and also what makes them unfit, on their own, to justify changing a production default.

So the recommended settings were run through **FlightListProcessor.process_dai** itself and scored the same way. Three differences between harness and pipeline are known in advance and are not bugs:

1. the pipeline selects among candidate aerodromes by **H3 ring count** (**distance_from_center**, an integer stepping about 5.2 km at resolution 7), while the harness uses exact haversine distance;
2. the pipeline filters zones by radius **band**, the harness by exact radius;

- the harness smooths barometric altitude across the flight-level boundary and the pipeline does not.

The gap those produce is the fidelity of every tuned number on this page.

Run	Role	Mode	Coverage	Accuracy	Correct	Wrong	Score
legacy	ADEP	trend	68.72%	98.82%	64,593	772	63,049
legacy	ADES	trend	68.06%	97.87%	63,362	1,376	60,610
trend	ADEP	trend	78.06%	98.48%	73,117	1,126	70,865
trend	ADES	trend	72.17%	94.38%	64,786	3,859	57,068
endpoint	ADEP	endpoint	80.96%	98.16%	75,587	1,416	72,755
endpoint	ADES	endpoint	69.10%	95.63%	62,856	2,870	57,116
nearest	ADEP	nearest	91.68%	87.78%	76,545	10,656	55,233
nearest	ADES	nearest	88.56%	79.41%	66,892	17,344	32,204
combined	ADEP	endpoint	80.90%	98.23%	75,585	1,365	72,855
combined	ADES	trend	68.26%	97.96%	63,602	1,325	60,952
recommended	ADEP	endpoint	80.90%	98.22%	75,585	1,368	72,849
recommended	ADES	trend	66.29%	98.24%	61,946	1,111	59,724

Harness against pipeline, at the same parameters:

Role	Method	Harness score	Pipeline score	Residual	Relative
ADEP	endpoint	72,755	72,755	+0	+0.00%
ADES	trend	61,038	60,952	−86	−0.14%

The two rows could hardly be more different, and the difference is the single most important result on this page.

endpoint reproduces exactly. That is not luck: the endpoint sweep filters the same cached candidate table the pipeline reads, applying the same rule, so harness and pipeline are the same computation and agree to the flight.

trend does not. The harness ranks candidate aerodromes by exact haversine distance. The pipeline first keeps only the candidates at the minimum H3 *ring count* — an integer stepping about 5.2 km at resolution 7 — and applies the haversine tie-break, and the scheduled-service penalty, only within that surviving set. An aerodrome one ring further out is gone before the penalty can promote it. So a penalty tuned against exact distance does something weaker in production, and the harness’s trend optimum is optimistic by the margin shown.

9.1 What running the pipeline caught

The first pipeline path run produced a result that no amount of sweeping could have: the scheduled-service penalty step scored **exactly** what the step before it did, to the flight. An identical score is not a weak effect, it is no effect, and a parameter with no effect is a parameter that is not reaching the code.

It was not. The penalty was added to the ranking in `_compute_flight_table`, but the projection a few lines above it dropped the `apt_scheduled` column, so the ranking looked the column up, failed to find it, and fell back to raw distance. A penalty of ten nautical miles and a penalty of zero produced byte-identical flight lists.

Two things made this visible rather than invisible. The pipeline run put the parameter in front of real data, where a sweep harness — which reads a *different* table that does carry the column — could not. And the fallback printed a note saying the column was missing instead of quietly ranking on distance, because a missing column and a zero penalty look identical in the output. The numbers below are from after the fix.

9.2 The trend tuning re-walked inside the pipeline

The path below runs through `process_dai` itself: production constants, then one tuned value adopted per step. It was run twice — once with the pipeline’s original **H3 ring-count** selection, and once with **exact-distance** ranking. The two disagree in sign, which is the whole finding.

Step	Ring: score	Ring: gain	Exact: score	Exact: gain	Coverage	Accuracy
production constants	60,610	—	60,265	—	68.06%	97.70%
+ scheduled-service penalty	60,844	+234	61,114	+849	68.06%	98.13%
+ flight-level cap	59,103	−1,741	61,776	+662	69.92%	97.63%
+ vote margin	59,223	+120	61,929	+153	70.17%	97.60%
+ radius	59,408	+185	62,018	+89	69.83%	97.79%

Under ring selection the tuning is worth −1,202; under exact distance it is worth +1,753. The flight-level cap is the step that flips hardest — −1,741 against +662 — and it is the step that admits samples from higher up, where more aerodromes share a ring and the ring rule is least able to separate them.

So the sweep harness was never over-estimating. It was measuring a *better algorithm* — one that measures every candidate before choosing — and the pipeline has now adopted it. `trend_rank_by` defaults to "haversine"; `DetectionConfig.legacy()` still returns "ring", so every published flight list stays reproducible.

9.3 Production’s own optimum

The path adopts the values the *harness* preferred. Whether those are also the **pipeline’s** optima is a separate question, and it is answered by sweeping the flight-level cap and radius through `process_dai` directly.

Score by flight-level cap (rows) and radius (columns), through the pipeline, at vote margin 2 and penalty 10 NM. Departures are shown for completeness – they are served by `endpoint`, so that half is a check rather than a recommendation.

FL cap	ADES 20 NM	ADES 30 NM	ADEP 20 NM	ADEP 30 NM
40	61,558	61,323	64,340	64,088
60	62,018	61,929	67,484	67,243
80	61,139	61,335	69,104	68,872
100	59,007	59,507	70,342	70,139
120	57,131	57,654	70,913	70,837

Arrivals peak at **FL60 / 20 NM** and fall away on both sides — an interior optimum, not an edge of the grid. Departures climb monotonically to **FL120**, but departures are served by **endpoint**, so that column is a check rather than a recommendation. Both roles prefer the tighter radius at every flight level.

These are the values that ship. Both had been rejected earlier in this study on ring-ranked evidence, and both are reinstated here on the pipeline’s own measurement.

One caveat, stated rather than buried: the grid holds the **vote margin** fixed at 2 while varying the cap and radius. That value therefore rests on the sweep harness alone, and is the weakest-supported of the shipped parameters.

10 Why each value, and not its neighbour

A recommendation is only as good as the evidence that picked it, so every value the sweeps select appears below with the score at that value, the score at the adjacent grid points, and the score at whatever production uses today. The last column is the one that matters most for trusting the number: a value on a broad **plateau** is a safe recommendation even from a single sample, because its neighbours score nearly the same and a little noise cannot move it far. A value on a sharp isolated **peak** may be partly the sample’s noise, and is flagged as such however good its score.

! These are the sweeps’ choices, not the shipped ones

The **Chosen** column below is the **harness argmax**, which is what this section exists to scrutinise. For **trend**, Section 9 is what settles whether each value survives in production — and under exact-distance ranking they do, which is why the harness numbers below and the shipped values in Section 15 now agree. They are still kept apart, because for a period they did *not* agree, and a sweep preference is only ever evidence until the pipeline confirms it.

Role	Parameter	Sweep		Score at	Score at	Neighbour	Neighbour	Δ vs	Shape
		Production	argmax	produc- tion	argmax	below	above	produc- tion	
ADES trend	FL cap	40	60	60,720	61,038	60,720	60,259	+318	peak
ADES trend	vote margin	4	2	60,843	61,038	60,996	60,843	+195	plateau
ADES trend	radius (NM)	30	20	61,023	61,038	59,740	61,023	+15	plateau
ADEP endpoint	radius (NM)	40	30	72,541	72,755	72,591	72,541	+214	plateau
ADEP endpoint	height (ft)	15000	15000	72,755	72,755	71,591	72,370	+0	peak

The penalty is treated separately because it is not a geometric parameter: it re-ranks candidates without changing how many are answered, so its sweep holds the geometry fixed and appears in Section 6.

10.1 Which change is actually doing the work

The table above varies one parameter at a time around the chosen setting, which shows whether each value is defensible but not how much each contributes to the total gain. This walks the whole distance instead: from the production constants to the recommended setting, changing one parameter per step.

	Step	Score	Gain from this step	Share of total gain
production constants (FL40, margin 4, 30 NM, no penalty)		59,322	—	—
+ scheduled-service penalty of 10 NM		60,216	+894	52%
+ flight-level cap FL60		60,830	+614	36%
+ vote margin 2		61,023	+193	11%
+ radius 20 NM		61,038	+15	1%

The gain is not spread evenly. **scheduled-service penalty of 10 NM** accounts for 52% of it on its own. That is worth stating plainly: the single most valuable change to arrival detection is not a tuned threshold at all, but applying to **trend** the scheduled-service tie-break that **endpoint** already had and **trend** never did.

By contrast the radius step is worth 15 flights out of 61,038 here — inside what one three-day sample can resolve, and on this evidence alone it would not justify moving a published constant. It is adopted anyway, on stronger evidence gathered later: the pipeline grid in Section 9 prefers the tighter radius at *every* flight level for *both* roles, which is a consistent effect across ten cells rather than a single argmax.

11 The verdict

Configuration	ADEPADES	ADEP cover-age	ADEP accu-racy	ADEP score	ADES cover-age	ADES accu-racy	ADES score	Total
SHIPPED endpoint + exact-distance trend	endpointtrend	80.90%	98.23%	72,855	68.26%	97.96%	60,952	133,807
control: production geometry + penalty, ring-ranked	endpointtrend	80.90%	98.22%	72,849	66.29%	98.24%	59,724	132,573
endpoint, both roles	endpointtrend	80.96%	98.16%	72,755	69.10%	95.63%	57,116	129,871
trend, departure parameters both roles	trend trend	78.06%	98.48%	70,865	72.17%	94.38%	57,068	127,933
legacy (production today)	trend trend	68.72%	98.82%	63,049	68.06%	97.87%	60,610	123,659
nearest (traffic library)	nearestnearest	91.68%	87.78%	55,233	88.56%	79.41%	32,204	87,437

The recommended configuration is the **combined** row: **endpoint** for departures, exact-distance **trend** for arrivals. It scores **133,807** against **production's 123,659**, a gain of **+10,148** on a three-day sample of 95,116 flights, and every figure in that table is a real pipeline run scored against Network Manager ground truth.

It is the shipped configuration in the strict sense: its trend cap, radius, vote margin, penalty and ranking rule, and its endpoint radius, height and penalty, were checked field by field against `DetectionConfig()`. The row is what production produces, not an approximation of it.

The **recommended** row is a **control, and a mislabelled one**. It was built by starting from `DetectionConfig.legacy()` and overriding the fields the author remembered to override – and `trend_rank_by` was not among them, so it kept ring-count selection and scored +8,914 against production while every properly-ranked configuration gained. It is left in the table because it is the cleanest demonstration on this page that the ranking rule matters more than the thresholds: identical geometry, one different selection rule, +1,234 of score.

11.1 What is being recommended, and why

Departures — endpoint, 30 NM, 15,000 ft, 10 NM scheduled-service penalty. Score 72,855 against production’s 63,049, a gain of +9,806. The clearest result in the study: an interior optimum in both parameters, better than `trend` at every exchange rate tried, and the pipeline reproduces the sweep exactly – the endpoint sweep filters the same cached candidate table the pipeline reads, so the tuned figure *is* the production figure.

Arrivals — trend at FL60, vote margin 2, 20 NM, 10 NM penalty, ranked on exact distance. Measured as an algorithm, on a trend-only flight list, that is 62,018 against production’s 60,610 – a gain of +1,408, won by converting 1,592 silences into correct answers for 92 extra wrong ones.

Inside the combined flight list the arrival half reads 60,952, +342 against production rather than the +1,408 just quoted. The difference is not the algorithm. Taking departures from `endpoint` adds tracks `trend` never produced, and where two share an aircraft, callsign and date, the scoring alignment awards the flight to whichever starts closest to the actual off-block time – sometimes now a departure-only track carrying no arrival. Accuracy *rises* in the merged table (97.96% against 97.87%) while coverage falls, which is the signature of lost pairings rather than wrong names.

One weak link, stated plainly. The vote margin of 2 comes from the sweep harness; the pipeline grid held it fixed while varying the cap and radius. Every other shipped value was measured through `process_dai`. If one parameter in this recommendation is going to fail on a fresh sample, that is the one.

11.2 The traffic library’s rule, for reference

nearest — the unconditional rule the `traffic` library uses — answers far more often than anything else here (91.68% of departures, 88.56% of arrivals) and is right far less often (87.78% and 79.41%). Scored, it is last by a wide margin: 87,437 against the recommendation’s 132,573.

That is not a criticism of `traffic`, which is solving a different problem: it names the nearest aerodrome and leaves the judgement to the caller. It is, however, the measurement that justifies OPDI carrying its own rule rather than adopting that one.

12 By aerodrome class

The candidate set is large and medium aerodromes only, so a flight whose true aerodrome is a small airfield or a heliport cannot be named correctly by any method on this page. This is the size of that population, and of everything else.

Departures

Class	Flights in truth	Aerodromes	Answered	Correct	Coverage	Accuracy
large_airport	80,661	331	71,174	70,597	88.24%	99.19%
out of area	7,892	1	911	626	11.54%	68.72%
medium_airport	6,069	352	4,705	4,362	77.53%	92.71%
small_airport	361	76	132	0	36.57%	0.00%
heliport	133	65	28	0	21.05%	0.00%

Arrivals

Class	Flights in truth	Aerodromes	Answered	Correct	Coverage	Accuracy
large_airport	80,683	331	61,045	60,257	75.66%	98.71%
out of area	7,964	1	26	0	0.33%	0.00%
medium_airport	5,983	346	3,761	3,345	62.86%	88.94%
small_airport	352	70	84	0	23.86%	0.00%
heliport	134	58	11	0	8.21%	0.00%

Outside the candidate set — 3 classes, 16,836 flights across both roles — the method is working against a reference it was never given. Those flights are counted in the denominator throughout this page, so every coverage figure here already carries their cost.

13 The busiest 100 aerodromes

Movement counts are what OPDI is for, so the last check is per aerodrome rather than per flight: for each of the hundred busiest in the reference, how many movements the recommended methodology finds, against how many the Network Manager data says there were, and how the algorithm production runs today compares on the same aerodromes.

Count ratio is movements found over movements in truth — the number a capacity or delay analysis would be distorted by. **Recall** is the share of that aerodrome’s true movements correctly attributed to it.

13.1 Departures

#	Aerodrome	Movements (truth)	Found	Correct	Count ratio	Recall	Recall, production	Δ recall
1	LTFM	2,282	2,205	2,198	0.967	96.32%	66.70%	+29.62 pp
2	EHAM	2,148	2,143	2,143	1.005	99.77%	99.81%	-0.05 pp

#	Aerodrome	Movements (truth)	Found	Correct	Count ratio	Recall	Recall, production	Δ recall
3	EDDF	2,094	2,082	2,082	0.995	99.43%	96.94%	+2.48 PP
4	LFPG	2,084	2,064	2,060	0.993	98.85%	99.04%	-0.19 PP
5	EGLL	2,001	1,989	1,988	1.011	99.35%	99.25%	+0.10 PP
6	LEMD	1,773	1,760	1,759	0.992	99.21%	94.59%	+4.62 PP
7	LEBL	1,591	1,580	1,579	0.992	99.25%	99.43%	-0.19 PP
8	EDDM	1,573	1,564	1,564	0.996	99.43%	99.36%	+0.06 PP
9	LIRF	1,449	1,444	1,444	0.997	99.65%	99.72%	-0.07 PP
10	LTAI	1,435	1,376	1,375	0.963	95.82%	61.25%	+34.56 PP
11	LEPA	1,417	1,400	1,400	0.989	98.80%	92.59%	+6.21 PP
12	LGAV	1,306	1,239	1,239	0.956	94.87%	23.28%	+71.59 PP
13	EGKK	1,223	1,216	1,216	0.996	99.43%	99.59%	-0.16 PP
14	EIDW	1,185	1,164	1,163	0.981	98.14%	98.23%	-0.08 PP
15	LSZH	1,178	1,171	1,164	0.991	98.81%	98.47%	+0.34 PP
16	LOWW	1,156	1,135	1,134	0.988	98.10%	98.88%	-0.78 PP
17	EKCH	1,140	1,132	1,132	0.994	99.30%	98.07%	+1.23 PP
18	LTFJ	1,137	1,102	1,101	0.980	96.83%	93.58%	+3.25 PP
19	LPPT	1,010	1,001	1,001	0.991	99.11%	80.00%	+19.11 PP
20	LFPO	992	982	982	0.997	98.99%	99.29%	-0.30 PP
21	LIMC	973	960	959	0.987	98.56%	99.28%	-0.72 PP
22	ENGM	932	922	922	0.995	98.93%	91.85%	+7.08 PP
23	EGCC	926	903	902	0.974	97.41%	99.35%	-1.94 PP
24	EGSS	885	882	882	0.997	99.66%	99.32%	+0.34 PP
25	EPWA	875	871	870	0.999	99.43%	99.20%	+0.23 PP
26	EBBR	872	869	869	1.009	99.66%	99.54%	+0.11 PP

#	Aerodrome	Movements (truth)	Found	Correct	Count ratio	Recall	Recall, production	Δ recall
27	EDDB	870	852	850	0.993	97.70%	97.47%	+0.23 pp
28	LEMG	850	847	847	0.999	99.65%	99.53%	+0.12 pp
29	ESSA	820	791	787	0.965	95.98%	96.34%	-0.37 pp
30	EDDL	758	746	744	0.989	98.15%	99.08%	-0.92 pp
31	LFMN	707	699	693	0.982	98.02%	97.74%	+0.28 pp
32	LSGG	680	676	676	1.010	99.41%	67.79%	+31.62 pp
33	LKPR	667	656	655	0.988	98.20%	98.65%	-0.45 pp
34	EFHK	659	653	653	1.002	99.09%	99.24%	-0.15 pp
35	LEAL	599	598	598	0.998	99.83%	99.83%	+0.00 pp
36	LHBP	578	555	553	0.958	95.67%	98.10%	-2.42 pp
37	EGPH	577	570	569	0.986	98.61%	98.96%	-0.35 pp
38	EGGW	550	549	549	1.002	99.82%	98.00%	+1.82 pp
39	EDDH	538	531	528	0.998	98.14%	98.33%	-0.19 pp
40	GCLP	520	23	0	0.000	0.00%	0.00%	+0.00 pp
41	EDDK	515	501	501	0.975	97.28%	96.89%	+0.39 pp
42	LLBG	513	501	501	0.992	97.66%	97.08%	+0.58 pp
43	LPPR	513	510	507	0.988	98.83%	99.03%	-0.19 pp
44	LROP	504	493	493	0.978	97.82%	98.02%	-0.20 pp
45	LEIB	495	490	490	0.990	98.99%	89.49%	+9.49 pp
46	EGBB	474	468	468	0.987	98.73%	98.52%	+0.21 pp
47	LIML	458	451	451	0.987	98.47%	98.47%	+0.00 pp
48	LIME	453	450	450	0.993	99.34%	98.68%	+0.66 pp
49	LIRN	449	381	335	0.751	74.61%	0.00%	+74.61 pp
50	LFML	444	440	439	0.989	98.87%	99.32%	-0.45 pp

#	Aerodrome	Movements (truth)	Found	Correct	Count ratio	Recall	Recall, production	Δ recall
51	EDDS	440	421	415	0.950	94.32%	93.41%	+0.91 pp
52	LGIR	424	6	0	0.000	0.00%	0.00%	+0.00 pp
53	LIPZ	412	408	408	0.993	99.03%	98.79%	+0.24 pp
54	LFLI	411	402	402	1.015	97.81%	98.78%	-0.97 pp
55	LPFR	405	401	399	0.985	98.52%	98.77%	-0.25 pp
56	LYBE	394	389	389	0.990	98.73%	98.22%	+0.51 pp
57	LICC	390	385	381	0.977	97.69%	0.00%	+97.69 pp
58	LTAC	381	7	0	0.000	0.00%	0.00%	+0.00 pp
59	LIPE	380	375	375	0.987	98.68%	98.42%	+0.26 pp
60	LCLK	368	353	351	0.957	95.38%	86.96%	+8.42 pp
61	GMMN	360	320	315	1.069	87.50%	62.78%	+24.72 pp
62	EPKK	360	349	347	0.967	96.39%	98.89%	-2.50 pp
63	EGGD	359	356	355	0.989	98.89%	99.44%	-0.56 pp
64	LEVC	355	342	341	0.961	96.06%	90.42%	+5.63 pp
65	LFSB	354	351	349	0.994	98.59%	98.31%	+0.28 pp
66	ENBR	349	343	343	1.026	98.28%	96.85%	+1.43 pp
67	LTBJ	346	324	275	0.795	79.48%	1.45%	+78.03 pp
68	EGPF	344	324	322	0.948	93.60%	93.60%	+0.00 pp
69	LGRP	340	327	326	0.976	95.88%	93.82%	+2.06 pp
70	LMML	326	324	324	0.997	99.39%	98.77%	+0.61 pp
71	GCTS	321	5	0	0.019	0.00%	0.00%	+0.00 pp
72	GCXO	319	306	306	1.141	95.92%	22.26%	+73.67 pp
73	LATI	316	7	0	0.000	0.00%	0.00%	+0.00 pp
74	LICJ	310	287	287	0.929	92.58%	53.87%	+38.71 pp

#	Aerodrome	Movements (truth)	Found	Correct	Count ratio	Recall	Recall, production	Δ recall
75	EBCI	304	304	304	1.000	100.00%	100.00%	+0.00 pp
76	BIKF	297	296	294	1.084	98.99%	98.32%	+0.67 pp
77	ELLX	294	292	292	0.993	99.32%	98.64%	+0.68 pp
78	LFBO	293	289	289	0.993	98.63%	98.98%	-0.34 pp
79	LGTS	279	271	270	0.971	96.77%	68.82%	+27.96 pp
80	GMMX	278	93	86	0.317	30.94%	0.36%	+30.58 pp
81	LEZL	275	274	273	0.993	99.27%	99.64%	-0.36 pp
82	LTBS	273	219	219	0.802	80.22%	0.00%	+80.22 pp
83	EVRA	273	271	270	0.993	98.90%	98.17%	+0.73 pp
84	LBSF	272	165	165	0.607	60.66%	0.00%	+60.66 pp
85	GCRR	269	1	0	0.000	0.00%	0.00%	+0.00 pp
86	EGNX	267	263	263	0.985	98.50%	98.88%	-0.37 pp
87	EPKT	259	251	249	0.961	96.14%	93.44%	+2.70 pp
88	HEGN	257	0	0	0.000	0.00%	0.00%	+0.00 pp
89	UGTB	257	165	159	0.774	61.87%	0.00%	+61.87 pp
90	LFRS	256	253	253	0.992	98.83%	78.52%	+20.31 pp
91	EDDP	256	255	255	1.004	99.61%	98.05%	+1.56 pp
92	EPGD	248	54	39	0.157	15.73%	6.45%	+9.27 pp
93	LGKR	244	209	207	0.869	84.84%	0.00%	+84.84 pp
94	HECA	244	62	42	0.172	17.21%	6.97%	+10.25 pp
95	EGAA	239	239	239	1.004	100.00%	76.15%	+23.85 pp
96	ENZV	236	163	160	0.737	67.80%	30.51%	+37.29 pp
97	LEBB	234	183	172	0.744	73.50%	0.00%	+73.50 pp
98	LFPB	230	227	227	1.000	98.70%	95.65%	+3.04 pp

#	Aerodrome	Movements (truth)	Found	Correct	Count ratio	Recall	Recall, production	Δ recall
99	LIBD	228	1	0	0.000	0.00%	0.00%	+0.00 pp
100	LIEO	228	201	194	0.851	85.09%	8.33%	+76.75 pp

13.2 Arrivals

#	Aerodrome	Movements (truth)	Found	Correct	Count ratio	Recall	Recall, production	Δ recall
1	LTFM	2,258	1,815	1,768	0.783	78.30%	71.30%	+7.00 pp
2	EHAM	2,136	1,877	1,871	0.876	87.59%	90.17%	-2.57 pp
3	EDDF	2,091	1,776	1,772	0.848	84.74%	87.61%	-2.87 pp
4	LFPG	2,066	1,713	1,711	0.850	82.82%	85.77%	-2.95 pp
5	EGLL	1,986	1,581	1,447	0.794	72.86%	85.45%	-12.59 pp
6	LEMD	1,786	1,535	1,533	0.861	85.83%	90.31%	-4.48 pp
7	LEBL	1,597	1,499	1,496	0.937	93.68%	95.62%	-1.94 pp
8	EDDM	1,571	1,469	1,465	0.933	93.25%	94.33%	-1.08 pp
9	LTAI	1,437	941	941	0.657	65.48%	62.21%	+3.27 pp
10	LEPA	1,436	1,415	1,414	0.986	98.47%	98.05%	+0.42 pp
11	LIRF	1,433	1,255	1,253	0.880	87.44%	90.16%	-2.72 pp
12	LGAV	1,317	629	623	0.475	47.30%	12.53%	+34.78 pp
13	EGKK	1,220	1,100	1,100	0.902	90.16%	92.54%	-2.38 pp
14	EIDW	1,188	1,084	1,083	0.914	91.16%	93.27%	-2.10 pp
15	LSZH	1,160	1,052	1,050	0.914	90.52%	91.98%	-1.47 pp
16	EKCH	1,152	1,056	1,043	0.905	90.54%	91.93%	-1.39 pp
17	LOWW	1,145	1,048	1,044	0.914	91.18%	93.54%	-2.36 pp
18	LTFJ	1,138	992	983	0.877	86.38%	87.52%	-1.14 pp
19	LPPT	1,001	880	877	0.877	87.61%	90.51%	-2.90 pp

#	Aerodrome	Movements (truth)	Found	Correct	Count ratio	Recall	Recall, production	Δ recall
20	LFPO	991	940	939	0.949	94.75%	95.76%	-1.01
21	LIMC	967	856	851	0.881	88.00%	91.62%	-3.62
22	ENGM	940	890	890	0.953	94.68%	95.21%	-0.53
23	EGCC	924	821	820	0.889	88.74%	90.91%	-2.16
24	EGSS	888	834	829	0.937	93.36%	94.71%	-1.35
25	EPWA	880	783	751	0.853	85.34%	88.64%	-3.30
26	EBBR	871	767	764	0.884	87.72%	89.90%	-2.18
27	LEMG	861	834	834	0.969	96.86%	98.03%	-1.16
28	ESSA	841	727	724	0.870	86.09%	86.21%	-0.12
29	EDDB	840	789	784	0.937	93.33%	94.52%	-1.19
30	EDDL	751	715	714	0.959	95.07%	97.20%	-2.13
31	LFMN	710	684	680	1.008	95.77%	96.48%	-0.70
32	LSGG	670	637	634	0.954	94.63%	96.72%	-2.09
33	EFHK	666	574	573	0.865	86.04%	86.64%	-0.60
34	LKPR	657	605	602	0.919	91.63%	93.76%	-2.13
35	LEAL	599	589	588	0.982	98.16%	98.83%	-0.67
36	EGPH	584	541	541	0.933	92.64%	94.69%	-2.05
37	LHBP	577	518	515	0.898	89.25%	93.59%	-4.33
38	EGGW	552	529	521	0.947	94.38%	95.83%	-1.45
39	EDDH	532	509	507	0.970	95.30%	97.93%	-2.63
40	GCLP	521	0	0	0.000	0.00%	0.00%	+0.00
41	LLBG	516	428	422	0.820	81.78%	88.37%	-6.59
42	LPPR	512	479	478	0.934	93.36%	93.75%	-0.39
43	LEIB	509	498	498	0.980	97.84%	97.64%	+0.20

#	Aerodrome	Movements (truth)	Found	Correct	Count ratio	Recall	Recall, production	Δ recall
44	EDDK	509	473	470	0.923	92.34%	95.09%	-2.75 pp
45	LROP	509	486	484	0.953	95.09%	97.25%	-2.16 pp
46	EGBB	466	419	418	0.897	89.70%	92.70%	-3.00 pp
47	LIML	465	457	455	0.987	97.85%	97.63%	+0.22 pp
48	LIME	460	447	444	0.965	96.52%	97.83%	-1.30 pp
49	LIRN	447	1	0	0.000	0.00%	0.00%	+0.00 pp
50	LFML	445	439	437	0.991	98.20%	99.10%	-0.90 pp
51	EDDS	433	409	409	0.958	94.46%	95.61%	-1.15 pp
52	LGIR	430	0	0	0.000	0.00%	0.00%	+0.00 pp
53	LFLI	412	398	397	0.998	96.36%	97.33%	-0.97 pp
54	LIPZ	409	388	386	0.946	94.38%	95.35%	-0.98 pp
55	LPFR	407	403	402	0.993	98.77%	98.77%	+0.00 pp
56	LYBE	395	364	364	0.922	92.15%	94.68%	-2.53 pp
57	LICC	391	344	327	0.836	83.63%	5.88%	+77.75 pp
58	LIPE	376	370	369	0.981	98.14%	98.94%	-0.80 pp
59	LTAC	374	0	0	0.000	0.00%	0.00%	+0.00 pp
60	LCLK	367	349	348	0.959	94.82%	97.00%	-2.18 pp
61	EGGD	360	334	334	0.933	92.78%	95.28%	-2.50 pp
62	LEVC	357	348	348	0.978	97.48%	98.04%	-0.56 pp
63	GMMN	357	265	265	0.751	74.23%	78.15%	-3.92 pp
64	ENBR	357	341	329	0.944	92.16%	92.16%	+0.00 pp
65	EPKK	356	342	332	0.935	93.26%	94.66%	-1.40 pp
66	LFSB	352	341	341	0.972	96.88%	97.73%	-0.85 pp
67	LGRP	346	316	315	0.913	91.04%	91.04%	+0.00 pp

#	Aerodrome	Movements (truth)	Found	Correct	Count ratio	Recall	Recall, production	Δ recall
68	EGPF	346	301	300	0.876	86.71%	88.44%	-1.73
69	LTBJ	341	15	14	0.041	4.11%	0.00%	pp +4.11
70	LMML	326	318	316	0.969	96.93%	97.85%	pp -0.92
71	LATI	323	0	0	0.000	0.00%	0.00%	pp +0.00
72	GCTS	323	0	0	0.000	0.00%	0.00%	pp +0.00
73	GCXO	320	2	2	0.006	0.62%	0.00%	pp +0.62
74	LICJ	314	138	137	0.439	43.63%	38.54%	pp +5.10
75	EBCI	303	297	296	0.977	97.69%	98.68%	pp -0.99
76	BIKF	296	90	89	0.304	30.07%	33.11%	pp -3.04
77	ELLX	294	271	271	0.925	92.18%	96.60%	pp -4.42
78	LFBO	290	287	285	0.983	98.28%	98.97%	pp -0.69
79	LGTS	281	265	265	0.943	94.31%	91.81%	pp +2.49
80	LTBS	280	8	8	0.029	2.86%	0.36%	pp +2.50
81	GMMX	279	14	13	0.047	4.66%	0.00%	pp +4.66
82	EVRA	278	227	226	0.817	81.29%	82.37%	pp -1.08
83	LEZL	276	264	263	0.957	95.29%	99.28%	pp -3.99
84	GCRR	269	0	0	0.000	0.00%	0.00%	pp +0.00
85	LBSF	267	3	1	0.004	0.37%	0.00%	pp +0.37
86	EDDP	261	236	235	0.904	90.04%	93.49%	pp -3.45
87	HEGN	258	0	0	0.000	0.00%	0.00%	pp +0.00
88	LFRS	257	251	250	0.977	97.28%	97.28%	pp +0.00
89	UGTB	257	48	47	0.187	18.29%	20.23%	pp -1.95
90	EPKT	257	242	239	0.930	93.00%	93.39%	pp -0.39
91	EGNX	255	231	231	0.906	90.59%	92.94%	pp -2.35
								pp

#	Aerodrome	Movements (truth)	Found	Correct	Count ratio	Recall	Recall, production	Δ recall
92	LIEO	252	111	110	0.440	43.65%	21.43%	+22.22 pp
93	LGKR	247	32	31	0.126	12.55%	0.00%	+12.55 pp
94	EGAA	246	234	149	0.606	60.57%	89.43%	-28.86 pp
95	EPGD	244	20	1	0.016	0.41%	0.41%	+0.00 pp
96	ENZV	236	71	68	0.305	28.81%	27.54%	+1.27 pp
97	LEBB	235	0	0	0.000	0.00%	0.00%	+0.00 pp
98	LTFE	232	199	199	0.879	85.78%	78.02%	+7.76 pp
99	HECA	231	75	62	0.268	26.84%	2.16%	+24.68 pp
100	LIBD	229	1	0	0.000	0.00%	0.00%	+0.00 pp

13.3 Across the hundred

Role	Movements in truth	Recall, shipped	Recall, production	Aerodromes improved	Median count ratio
Departures	65,382	92.37%	83.40%	61 of 100	0.988
Arrivals	65,360	79.58%	79.75%	19 of 100	0.903

A count ratio below one means the aerodrome is under-counted — the usual case, since every abstention is a movement not attributed. Above one means movements are being attributed to an aerodrome that did not see them, which is the more damaging error and the one the scoring penalises at rate k .

The departure row is the result: recall rises by nearly nine points and the median aerodrome is within about one percent of its true movement count.

The arrival row must be read with Section 11 in hand. Recall reads 79.58% against production’s 79.75%, improving on 19 of 100 — which looks like a mild regression and is not one. Measured as an algorithm on a trend-only flight list, arrivals gain; the shortfall here comes from the merged list also containing departure-only tracks, which this table pairs with reference flights by start time, so some arrivals are counted against a track that carries none. The measurement moved; the data did not. It is left in the table rather than corrected away, because a benchmark that quietly adjusts its own alignment to flatter a result is worth less than one that shows where it is fragile.

14 Does any of it hold on a second period?

Everything above is one three-day sample. An argmax over hundreds of cells on a single sample always flatters itself: part of the winning cell’s margin is that sample’s noise, and noise does not

reproduce. The only way to find out which part is real is to measure a different period with the same code.

The second period is **5–7 June 2024**, chosen because the Network Manager reference for that month is committed to the repository and the matching tracks already exist from earlier versions of this study.

The 2024 sample carries 92,799 ground-truth flights against 2025’s 95,116.

Role	Parameter	2025 optimum	2024 optimum	Agrees
ADEP	trend FL cap	120	100	no
ADEP	trend vote margin	0	0	yes
ADEP	trend radius (NM)	20	20	yes
ADES	trend FL cap	60	80	no
ADES	trend vote margin	2	2	yes
ADES	trend radius (NM)	20	30	no

3 of 6 parameter optima agree across the two periods.

On the arrival side, where this study rejected the sweep’s preferences, the 2024 sweep prefers FL80 against 2025’s FL60 — and its own gain over the production constants is +5,063, against +1,716 on 2025. Both are harness figures, and Section 9 showed harness gains of this size on **trend** do not survive the pipeline’s ranking. A parameter that is neither stable across periods nor reproducible in production is not one to ship, which is what the defaults reflect.

The endpoint rule was not re-swept on the second period. Its sweep runs over the cached candidate table, which exists for 2025 and not for 2024; building it for 2024 would mean either materialising an H3 index over the whole second period or reimplementing the pipeline’s endpoint extraction in the harness. The second option is precisely the kind of shortcut Section 9 exists to warn about, so neither was taken.

The departure recommendation therefore rests on a different guarantee, and a stronger one for this particular purpose: it was validated *in the pipeline itself* on 2025, where the sweep and **process_dai** agree to the flight. What is missing is a second sample, not a second implementation. It is listed in the limitations as the outstanding check.

15 What ships

The pipeline’s **DetectionConfig** defaults are now the values this study measured. Six changed. The ordering below is deliberate: the ranking rule comes first, because without it none of the rest holds.

Parameter	Production	Now ships	Evidence
trend_rank_by	ring	haversine	The enabling change. Ring-count selection discarded candidates before measuring them; every tuned trend value fails under it and gains under exact distance.

Parameter	Production	Now ships	Evidence
<code>trend_max_fl</code>	40	60	Pipeline grid over FL 40–120: arrivals peak at FL60 and fall on both sides.
<code>trend_radius_nm</code>	30	20	Preferred at every flight level in the grid, both roles.
<code>trend_vote_margin</code>	4	2	Sweep harness only — the grid held it fixed. The weakest-supported value here.
<code>trend_sched_penalty_nm</code>	10	10	A tie-break <code>trend</code> never had. Improves 9 of the hundred busiest aerodromes, worsens none.
<code>endpoint_radius_nm</code>	40	30	Interior optimum; pipeline reproduces the sweep exactly.
<code>endpoint_height_ft</code>	15,000	15,000	Confirmed where it already was.
<code>endpoint_sched_penalty_nm</code>	10	10	Unchanged.

Three of these — the cap, the radius and the vote margin — were **rejected earlier in this study and reinstated**. The rejection was measured under ring-count selection, where raising the cap genuinely did lose ground. The lesson is not that the first measurement was careless but that it answered a narrower question than it appeared to: *given this selection rule*, the tuning fails. Changing the rule changed the answer.

Every OPDI flight list published before this change was built with the production column, and `DetectionConfig.legacy()` still returns exactly those values — including `trend_rank_by = "ring"` — so anything already released stays reproducible by name. That is the same principle as the frozen `version` strings on published events. A regression test pins the preset, a second pins the shipped values, and a third asserts the two differ, so the preset cannot quietly become vacuous.

 Re-running a past month will not reproduce it

These defaults change ADEP and ADES for any month re-processed with them. That is intended, and it is why the old behaviour remains reachable by name rather than being deleted. A new algorithm gets a new `version`; the published one is never mutated.

16 Limitations

16.1 Parameters that were not swept

Five parameters were held fixed by decision rather than by evidence. Naming them matters: a reader who sees seven tuned values and no caveat will reasonably assume the whole configuration was optimised, and it was not.

Parameter	Where	Held at	Why not swept
Smoothing half-window	<code>flights.py</code>	2 samples	Parameterised in config so it is visible and changeable, but a smoothing width interacts with the sample rate rather than the geometry, and the decimation study settled the sample rate separately.
Classification dead band	<code>flights.py</code>	none exists	There is no dead band to sweep. The vote margin already plays that role.
Out-of-area border margin	<code>flights.py</code>	30 NM	Fixes what counts as “outside the observed area”. Changing it redefines the ground truth rather than the method, so a sweep would move the target as well as the aim.
Aerodrome type set	<code>h3_airport_zones.py</code>	large + medium	The candidate set. Widening it is a data-volume decision about the zone table, not a detection threshold.
H3 resolution	<code>config.py</code>	7 (± 0.76 NM)	Changing it invalidates every cached zone table and the <code>distance_from_center</code> ring counts that production selects on.

Two of these carry a cost that this study measures rather than waves at, since “not swept” is not the same as “no consequence”.

The type set. 980 flights across both roles have a true aerodrome outside the large/medium candidate set (267 at heliports, 713 at small airports). No method on this page can name them, whatever its thresholds, because the aerodrome is not among the candidates. They are counted in every denominator here, so each coverage figure already carries their cost: they are worth about 0.52% of headline coverage.

The border margin. Fixed at 30 NM, it decides which flights are labelled out-of-area. 8.30% of ground-truth departures genuinely begin outside the observed area, and the recommended configuration recognises 7.93% of them — at 89.43% precision, so the label is trustworthy when it appears but is rarely reached. For arrivals the recall is 0.00%: **trend** emits no out-of-area label at all, which is a structural property of the vote rule rather than a tuning choice.

Both costs fall on coverage rather than accuracy, which is the benign direction: with the free out-of-area wins removed, in-area accuracy is 98.58% for departures and 98.27% for arrivals.

16.2 The outstanding check

The endpoint radius and height — the values that actually change what the pipeline ships for departures — have been validated against the pipeline but on **one** period. **trend**, whose tuned values were mostly rejected, is the one measured on two. That is the wrong way round, and it is the first thing a follow-up should fix: build the endpoint candidate cache for a second period and re-run the radius x height grid against it.

Two things limit how much that weakens the recommendation. The departure gain is large — thousands of flights, not tens — so it is not the kind of margin a single sample’s noise produces. And the optimum sits on a plateau in both parameters, so even a moderate shift in the argmax would cost little.

16.3 One sample, and what that does and does not license

The parameters are chosen as the argmax over a grid evaluated on a single three-day period. An argmax over hundreds of cells on one sample will always overstate the gain slightly: some of the winning cell’s margin is the sample’s noise, and it will not reproduce.

That is the reason for the second period, and for reporting the shape of each optimum rather than only its location. A parameter sitting on a broad plateau is a safe recommendation even from one sample; one sitting on a sharp isolated peak is not, however good its score.